Competing Risks Analysis of MLB Draft Data

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Introduction and Motivation

- In 2021, Major League Baseball (MLB) realigned Minor League Baseball (MiLB) and cut the number of rounds in the June amateur draft from 40 to 20

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- Less than 20% of draftees play a single game in MLB. The recent changes are meant to:
  - make the pipeline from the draft to MLB more efficient
  - improve quality of life for MiLB players who play for many years in the minors

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- When the draft occurs, performance data that might be used to predict this success is limited
- Given recent drafts, we can describe three main outcomes/events for draftees:
  - reached the MLB (played at least one game; **Event 1**)
  - retired without ever reaching MLB (**Event 2**)
  - still playing in MiLB and may or may not reach MLB (**Censored Event**)

Question of Interest

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- Given **time to event data**, some of which are **censored**, and with **multiple competing events**, the natural approach for answering this question is competing risks analysis

Data

- All signed draftees from 2012 to 2016 ($n = 4573$)
- Target feature: Years until event
  - considered censored if they had not reached MLB or retired by 2022
- Final predictor features:
  - Position when drafted (C, IF, OF, LHP, RHP)
  - Overall draft pick number
  - Type (high school, junior college, 4-year college)
  - Bonus as a proportion of pick number slot value (i.e. how much draftee was given out of amount allotted to their pick position)
- All data collected manually from Baseball-Reference[9] and Baseball America[2]

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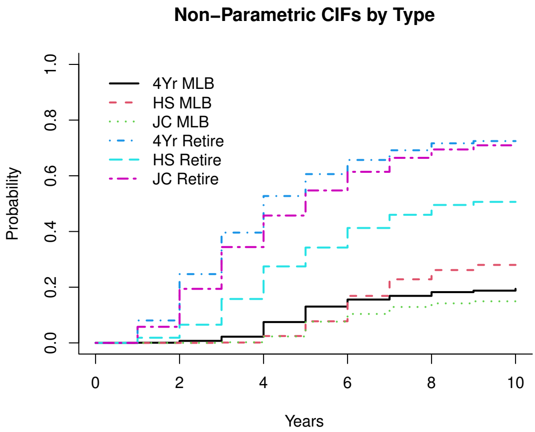
- Non-parametric approach (Nelson, 1969, 1972, Aalen, 1978)
- Proportional hazards approach (Fine and Gray, 1999)
- Accelerated failure time approach (Mahani and Sharabiani, 2019)

Non-Parametric Approach

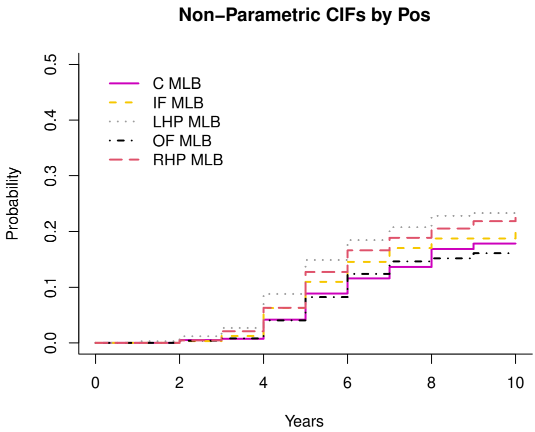
- Pros: Makes no assumptions on behavior of data, easily implemented, easily interpreted tests for equality across categorical groups (Gray, 1988)
- Cons: Incapable of handling continuous covariates, limited scope of inference (i.e. quantifying covariate effects)

Non-Parametric Results (Type)



- Reaching MLB: (Gray's 22.62, $p < .001$)
- Retiring: (Gray's 176.16, $p < .001$)

Non-Parametric Results (Position)



- Reaching MLB: Gray's 15.93, $p < .001$
- Retiring (not pictured): Gray's 7.40, $p = .116$

Non-Parametric Results (Takeaways)

- Evidence that some draft day factors likely do impact time to reach MLB or retiring without doing so
- High school draftees much less likely to retire over time, also less likely to reach MLB (until year 6)
- Pitchers more likely, especially LHP, to reach MLB over time
- Results seem to make intuitive sense, but approach cannot account for continuous features

Proportional Hazards Approach

- Pros: Can incorporate continuous covariates and complex interaction terms, parameter estimates allow for quantification of covariate effects (via odds ratios), simple to use model to predict individual player hazard functions
- Cons: Assumes the ratio between the covariate effects remains constant over time (i.e. the odds of survival between values are proportional at any time t)

Proportional Hazards Results (Reach MLB: 1)

Selected parameters, odds ratios (e^{β}), and odds ratio 95% confidence intervals for the Fine-Gray model assessing impact of features on time to reach MLB

Feature	β	e^{β}	95% CI LB	95% CI UB
Overall Pick (OvP)	-1.719	0.179	0.111	0.290
Bonus/Slot (BSp)	-0.002	0.998	0.778	1.280
HS	-0.139	0.870	0.506	1.498
JC	-0.239	0.787	0.505	1.228
IF	0.282	1.326	0.733	2.398
LHP	0.942	2.565	1.491	4.414
OF	-0.109	0.897	0.467	1.722
RHP	0.696	2.006	1.191	3.376

- Due to space constraints, the above table does not include the many significant interaction effects

Proportional Hazards Results (Retire: 2)

Selected parameters, odds ratios (e^{β}), and odds ratio 95% confidence intervals for the Fine-Gray model assessing impact of features on time to retire

Feature	β	e^{β}	95% CI LB	95% CI UB
Overall Pick (OvP)	0.481	1.617	1.510	1.733
Bonus/Slot (BSp)	-0.371	0.690	0.620	0.768
HS	-0.382	0.683	0.599	0.778
JC	-0.074	0.929	0.812	1.063
IF	-0.051	0.950	0.840	1.075
LHP	-0.169	0.844	0.738	0.966
OF	0.057	1.058	0.931	1.203
RHP	-0.156	0.855	0.763	0.959

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Proportional Hazards Results (Interpretation)

Example interpretations given baseline categories of Type=4Yr and Pos=C and average (512th) pick position with average bonus as a proportion of slot value:

- $e^{\beta(LHP,1)} = 2.565$: College left-handed pitchers are 156.5% more likely to reach MLB relative to catchers
- $e^{\beta(LHP,2)} = .844$: College left-handed pitchers are 15.6% less likely to retire before reaching MLB than catchers

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- **Note:** due to interaction effects, this is not true for different overall pick position values

Proportional Hazards Results (Takeaways)

- Allowing for continuous covariates and complex interactions allows for more robust (if sometimes tricky) inference
- Model is useful for making individual player predictions. Given a draftee's data, can input and produce full predicted risks for a player. For example:

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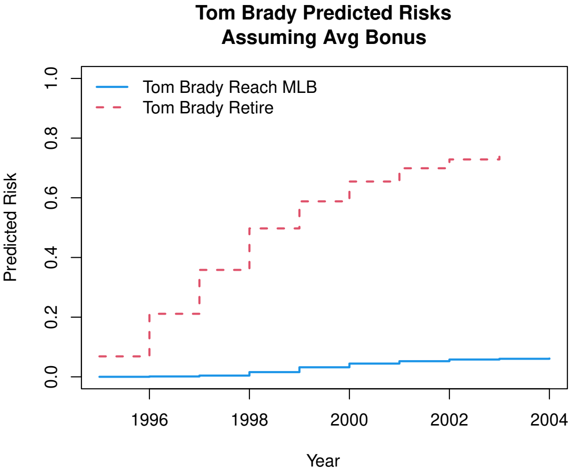
- Allowing for continuous covariates and complex interactions allows for more robust (if sometimes tricky) inference
- Model is useful for making individual player predictions. Given a draftee's data, can input and produce full predicted risks for a player. For example:
  - In the 1995 draft, Tom Brady was the 507th overall pick, out of High School, as a Catcher. While he didn't sign, we might assume he would have received the average bonus of $31,000 for his pick position against a slot value of $100,000:

Proportional Hazards Results (Takeaways)



Proportional Hazards Results (Drawbacks)

- However, there is evidence that the model is not totally appropriate due to potential violation of the proportional hazards assumption for Type:
  - $e^{\beta_{(HS,1)}} = .870$ implies that high school catchers are **always** 23% less likely to reach MLB than 4-year college catchers (given average pick position and bonus)
  - This contradicts non-parametric results from earlier

Accelerated Failure Time (AFT) Approach

- Pros: Allows for covariate effects to vary over time, covariate effects are still interpretable, allows for investigation of different distributions for survival function
- Cons: Does not allow for interaction effects between covariates, substantially longer computation time

AFT Results (Reach MLB: 1)

Parameters and time factors for the AFT model assessing impact of features on time to reach MLB

Feature	β	e^{β}
OvP	0.253	1.288
BSp	-0.030	0.970
HS	0.254	1.289
JC	0.159	1.173
IF	-0.026	0.974
LHP	-0.102	0.903
OF	0.053	1.054
RHP	-0.053	0.949

AFT Results (Retire: 2)

Parameters and time factors for the AFT model assessing impact of features on time to retire

Feature	β	e^{β}
OvP	-0.205	0.815
BSp	0.146	1.157
HS	0.230	1.259
JC	0.109	1.116
IF	0.026	1.026
LHP	0.068	1.071
OF	-0.024	0.976
RHP	0.088	1.092

AFT Results (Interpretation)

A unit increase in a covariate indicates the mean survival time will change by a factor of e^{β}

- a positive coefficient implies the covariate decelerates average event time as the covariate increases
- a negative coefficient implies the covariate accelerates the average event time as it increases

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Example:

- $e^{\beta_{OvP,1}} = 1.288$: implies that it takes longer for draftees to reach the MLB on average the later in the draft they are selected
- $e^{\beta_{OvP,2}} = 0.815$: implies that it takes less time for draftees to retire on average the later in the draft they are selected

AFT Results (Takeaways)

- Likelihood ratio test statistics for each cause (χ^2 values) suggest these models are useful for assessing survival:
  - Reach MLB: $\chi^2 = 405.1$, $p < .001$
  - Retire: $\chi^2 = 972.9$, $p < .001$

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  - Reach MLB: $\chi^2 = 405.1$, $p < .001$
  - Retire: $\chi^2 = 972.9$, $p < .001$
- Directly comparing AFT results to equivalent proportional hazards models shows general agreement in average coefficient effects, with slight deviations in effects over time

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  - High School draftees are generally less likely to retire over time than college players, and initially less likely to reach MLB though this changes over time (around year 6 in both Non-Parametric and AFT models)
  - Pitchers (esp. LHP) generally have greater chances to reach MLB over time and take longer to retire from MiLB than other positions, though this varies by pick position

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  - Pitchers (esp. LHP) generally have greater chances to reach MLB over time and take longer to retire from MiLB than other positions, though this varies by pick position
  - Draftees should strive to maximize their pick position and bonus amount, which likely increases team investment in player success

Summary and Future Work

- Next Steps:
  - Adjust to account for injury information
  - Consider including HS/college performance data
  - Find more computationally efficient ways to predict players using AFT model
  - Continue improving R Shiny app for interactive prediction

QUESTIONS?

- **email**: e.gerber@northeastern.edu
- QR Code to Shiny App:

References

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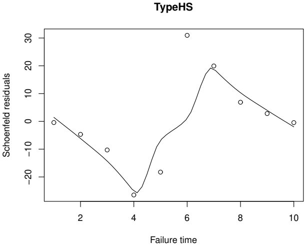
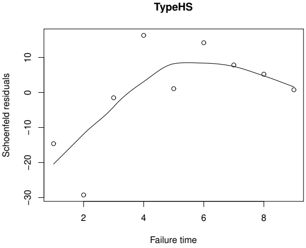
Proportional Hazards Results (Drawbacks Cont.)

We can also check Schoenfeld residuals (Schoenfeld, 1982). Plotted against survival time for each feature: if the proportional assumption for a given covariate is true, the residual plot should have a consistent mean over time:





- Note that while the Schoenfeld residuals for most other covariates look good, this suggested violation of the assumption leads to our final approach